Andrew Geyko

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Profile

PhD student in computer science at Saarland University, advised by Prof. André Brinkmann. I work on system-model co-design for machine learning, mostly where compute and memory are limited, such as on-device training and personalization, and KV-cache management for LLM inference. Before that I worked on scalable MPI communication (first author, IEEE HiPC 2024) and on the energy cost of distributed LLM training.

Education

10/2026 – Present
Ph.D. Computer Science, Saarland University, Saarbrücken, Germany.
Advisor: Prof. André Brinkmann.

10/2024 – 09/2026
CS@max planck graduate program
Saarland University, Saarbrücken, Germany.

08/2020 – 05/2024
B.S. Computer Science, University of New Mexico, Albuquerque, USA.
GPA: *4.03/4.33*, graduated *summa cum laude*
Includes an exchange semester at the University of Bergen, Norway (fall 2023)

Research Experience

08/2025 – 12/2025
Research Intern. *Max Planck Institute for Software Systems (group of Prof. Antoine Kaufmann), Saarbrücken, Germany.*

- Analyzed energy/performance trade-offs of distributed LLM training across GPU generations (V100/H100) using the ASTRA-sim simulator.
- Automated sweeps over 4D-parallelism strategies (data, tensor, sequence, pipeline) in Python; modeled per-GPU power draw to estimate energy consumption vs. time-to-completion.
- Containerized the full experiment pipeline with Docker for reproducibility.
- Code: `github.com/gordofreemo/energy-aware-ml-migration`

04/2025 – 06/2025
Research Intern. *Max Planck Institute for Security and Privacy (group of Dr. Clara Schneidewind), Bochum, Germany.*

- Developed soundness proofs for a smart-contract verification scheme.

01/2023 – 05/2024
Undergraduate Research Assistant. *Scalable Solvers Lab (group of Prof. Amanda Bienz), University of New Mexico, Albuquerque, USA.*

- Developed locality-aware algorithms for sparse dynamic data exchange in C++, CUDA, and MPI, achieving 128× speedups over standard approaches.
- Extended the algorithms to heterogeneous (GPU) architectures and benchmarked at scale.
- First-author publication at IEEE HiPC 2024.

Research Experience (continued)

06/2023 – 08/2023 **Undergraduate Complexity Researcher**. *Santa Fe Institute, Santa Fe, USA.*

- Developed a novel one-dimensional ConceptARC dataset for evaluating the abstract reasoning capabilities of LLMs under the supervision of Prof. Melanie Mitchell.
- Built a program-synthesis solver for the new dataset as part of the Undergraduate Complexity Research (UCR) program, extending a Kaggle-winning ARC solver.

Publications

A. Geyko, M. Mosbach, and A. Brinkmann, "Fine-Tuning a 3B-Parameter LLM on a Smartphone: Characterizing Sustained Training," arXiv preprint, 2026. ARXIV: 2610.06325

A. Geyko, G. Collom, D. Schafer, P. Bridges, and A. Bienz, "A More Scalable Sparse Dynamic Data Exchange," in *Proc. 31st IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC)*, Bengaluru, India, 2024, pp. 143–154. DOI: 10.1109/HiPC62374.2024.00023

Skills

Languages	English (native), Russian (native), German (B1)
ML & LLM	PyTorch, HuggingFace Transformers, PEFT/LoRA fine-tuning, quantization, MLX, RAG, Matplotlib/SciPy, agentic AI development workflows
Programming	Python, C, C++, Java, CUDA, Swift, Haskell, Scheme
Systems & Tools	Linux, Bash, Docker, Git, MPI, SLURM, HTCondor

Awards & Scholarships

2020 – 2024 **Dean's List**, University of New Mexico.

2022, 2023 **Bob Veroff Scholarship** (2022) and **Tallant Cooper Scholarship** (2023), University of New Mexico.

Saarbrücken, 20.10.2026